Question 1. DDPM: Deriving the Reverse Posterior

Consider the scalar forward diffusion process

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{\beta_t}\, z_t, \qquad z_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1),$$

where $0 < \beta_t < 1$, $\alpha_t = 1 - \beta_t$, and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. Each z_t is independent of $x_0, z_1, \ldots, z_{t-1}$. The index $t = 1, \ldots, T$ denotes a diffusion step, and $\mathcal{N}(x; \mu, v)$ denotes a Gaussian density with mean μ and variance v.

For $t \geq 2$, you may use

$$q(x_{t-1} \mid x_0) = \mathcal{N}\Big(x_{t-1}; \sqrt{\bar{\alpha}_{t-1}}\, x_0,\, 1 - \bar{\alpha}_{t-1}\Big).$$

(a) Posterior distribution. [25 points] Use Bayes' theorem and complete the square in x_{t-1} to show that

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t).$$

Derive explicit expressions for the posterior mean $\tilde{\mu}_t$ and variance $\tilde{\beta}_t$ in terms of $x_t, x_0, \alpha_t, \beta_t$, and $\bar{\alpha}_{t-1}$. Show the quadratic and linear coefficients in your derivation.

(b) Noise parameterization. [15 points] The marginal forward process can also be sampled using

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,1).$$

Eliminate x_0 from your result in part (a) and show that

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}}\, \varepsilon \right).$$

Explain how ε differs from the single-step noise z_t.

(c) Learning the reverse process. [10 points] During generation, x_0 and the realized forward noise are unavailable. A neural network $\varepsilon_\theta(x_t, t)$ predicts the accumulated noise. Write the resulting reverse mean $\mu_\theta(x_t, t)$ and the usual simplified mean squared error training objective, with x_0 sampled from the data distribution, $t \sim \text{Uniform}\{1, \ldots, T\}$, and $\varepsilon \sim \mathcal{N}(0,1)$ sampled independently. Explain why knowing the exact posterior during training does not eliminate the need to learn a reverse model.

Question 2. AR(1): Effective Sample Size

Let $Y_1, \ldots, Y_{10}$ be independent samples from $\mathcal{N}(0,1)$. Compare them with N consecutive observations from the Gaussian AR(1) process

$$X_n = \phi X_{n-1} + \sqrt{1-\phi^2}\, Z_n, \qquad Z_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad 0 \le \phi \le 1.$$

Initialize $X_0 \sim \mathcal{N}(0,1)$ independently of all innovations. Then $X_n \sim \mathcal{N}(0,1)$ and $\mathrm{Cov}(X_i, X_j) = \phi^{|i-j|}$.

Define the mean and variance estimators

$$\bar{X}_N = \frac{1}{N}\sum_{i=1}^{N} X_i, \qquad V_{X,N} = \frac{1}{N}\sum_{i=1}^{N} X_i^2,$$

with $\bar{Y}_{10}$ and $V_{Y,10}$ defined analogously. The population mean is **known to be zero**; therefore, do not subtract the sample mean when calculating V.

(a) Sampling uncertainty. [15 points] Derive the exact finite-N variances of $\bar{X}_N$ and $V_{X,N}$, and the corresponding variances for the independent reference. You may use

$$\mathrm{Cov}(U^2, W^2) = 2\big[\mathrm{Cov}(U,W)\big]^2$$

for jointly Gaussian, zero-mean variables U, W. By matching these variances to those obtained from independent observations, derive the effective sample size for the mean and the effective degrees of freedom for the variance estimator.

(b) Required sample sizes. [15 points] For each

$$\phi \in \{0,\ 0.1,\ 0.5,\ 0.8,\ 0.9,\ 1.0\},$$

find the smallest integer $N \ge 10$ satisfying

$$\mathrm{Var}(\widehat{\theta}_{X,N}) \le 1.01\ \mathrm{Var}(\widehat{\theta}_{Y,10}),$$

separately for the mean and variance estimators. Here, $\widehat{\theta}$ denotes the estimator being compared. This criterion allows at most 1% greater sampling variance than the independent reference and also accepts lower sampling variance.

Plot the two threshold sample sizes against ϕ. Compare them with the large-N approximations obtained by extending the autocorrelation sums to infinity. Do not restrict the search to $N \le 120$. If no finite threshold exists, explain why and indicate this explicitly on the figure.

(c) Monte Carlo comparison. [20 points] For each ϕ with finite thresholds, use at least 20,000 independent Monte Carlo repetitions to compare the sampling distributions of each estimator for: (i) 10 independent observations, (ii) 10 consecutive AR(1) observations, and (iii) the corresponding threshold sample size from part (b).

Plot normalized histograms using common bins within each comparison, and report the simulated expectations and variances. Use theory, rather than simulation noise, to determine the thresholds.

Explain why the mean and variance estimators require different sample sizes, and why comparable sampling variances do not generally imply identical sampling distributions. For $\phi = 1$, describe the process and explain whether increasing N reduces either estimator's uncertainty.